

**SLEIPNER A OIL PLATFORM, Norway**WMO: **010886**Lat: **58.367N**Lon: **1.900E**Elev: **98**StdP: **14.64**Time Zone: **1.00 (EUC)**Period: **10-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 31.9       | 34.1 | 19.5                          | 14.7 | 34.8 | 22.8 | 17.2 | 37.2 | 68.1                  | 41.1 | 64.8 | 40.6 | 29.9                  | 110  | 1.224 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.1                    | 63.0            | 60.3 | 62.4 | 59.8 | 61.0 | 58.7 | 61.5                | 63.4 | 60.2 | 61.8 | 59.4 | 61.0 | 24.6                 | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 60.9                            | 80.0 | 62.4 | 59.3 | 75.7 | 61.1 | 58.8 | 74.2 | 60.6 | 27.4          | 62.7 | 26.6 | 62.6 | 25.9 | 60.3 | 66.7              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 61.8          | 55.9 | 50.7 | DB                         | 31.0 | 65.2               | 4.1 | 2.7                                                | 28.0 | 67.2       | 25.6 | 68.7       | 23.3 | 70.3       | 20.3 | 72.2 | (4) |
| (5)               |      |      | WB                         | 27.9 | 62.2               | 4.4 | 2.3                                                | 24.8 | 63.9       | 22.2 | 65.3       | 19.8 | 66.6       | 16.6 | 68.3 | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 48.8   | 42.4  | 41.9  | 42.2  | 44.2  | 48.3  | 52.9  | 57.1  | 58.2  | 55.6  | 51.2  | 47.3  | 44.3 |
|                                                                                  | DBStd       | 6.55   | 3.40  | 3.46  | 3.81  | 2.71  | 3.28  | 2.28  | 2.94  | 2.05  | 2.42  | 2.76  | 2.94  | 3.97 |
|                                                                                  | HDD50       | 1240   | 236   | 226   | 242   | 175   | 73    | 4     | 0     | 0     | 0     | 19    | 89    | 176  |
|                                                                                  | HDD65       | 5896   | 701   | 646   | 706   | 625   | 518   | 364   | 245   | 211   | 281   | 427   | 532   | 640  |
|                                                                                  | CDD50       | 819    | 0     | 0     | 0     | 0     | 20    | 90    | 221   | 254   | 169   | 57    | 7     | 1    |
|                                                                                  | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 26.9   | 33.2  | 30.9  | 28.1  | 25.0  | 23.8  | 21.5  | 20.9  | 22.1  | 25.9  | 29.1  | 29.9  | 33.0 |
| Precipitation                                                                    | PrecAvg     | 49.4   | 5.9   | 5.0   | 4.6   | 2.5   | 1.9   | 2.5   | 3.0   | 3.6   | 4.4   | 5.0   | 5.3   | 5.7  |
|                                                                                  | PrecMax     | 61.4   | 8.5   | 10.4  | 8.2   | 4.1   | 3.6   | 4.1   | 5.3   | 9.1   | 6.5   | 8.8   | 7.2   | 11.4 |
|                                                                                  | PrecMin     | 38.0   | 2.4   | 2.4   | 2.2   | 1.7   | 0.8   | 1.1   | 1.3   | 1.5   | 1.6   | 1.9   | 2.5   | 2.9  |
|                                                                                  | PrecStd     | 6.0    | 1.9   | 1.9   | 1.5   | 0.7   | 0.6   | 0.7   | 0.9   | 1.3   | 1.3   | 1.6   | 1.3   | 1.7  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 50.1  | 49.8  | 50.2  | 52.1  | 61.1  | 59.2  | 68.0  | 64.3  | 62.6  | 57.6  | 54.0  | 52.2 |
|                                                                                  |             | MCWB   | 49.0  | 48.3  | 47.3  | 47.1  | 57.1  | 55.7  | 63.7  | 61.6  | 60.6  | 55.5  | 52.0  | 50.2 |
|                                                                                  | 2%          | DB     | 48.5  | 48.2  | 48.3  | 50.1  | 57.4  | 57.5  | 64.5  | 62.8  | 61.0  | 57.0  | 53.4  | 51.5 |
|                                                                                  |             | MCWB   | 47.4  | 46.6  | 46.9  | 46.4  | 53.4  | 54.8  | 61.7  | 60.2  | 59.2  | 54.8  | 51.6  | 49.7 |
|                                                                                  | 5%          | DB     | 48.2  | 46.8  | 46.8  | 48.5  | 54.0  | 57.1  | 62.8  | 62.2  | 59.4  | 55.6  | 52.0  | 50.2 |
|                                                                                  |             | MCWB   | 47.0  | 45.4  | 45.3  | 46.2  | 50.8  | 54.5  | 60.2  | 59.6  | 57.2  | 52.8  | 50.3  | 48.6 |
|                                                                                  | 10%         | DB     | 46.7  | 46.4  | 46.5  | 47.9  | 52.1  | 55.7  | 61.1  | 61.0  | 59.0  | 55.0  | 51.6  | 49.6 |
|                                                                                  |             | MCWB   | 45.0  | 45.1  | 45.0  | 46.0  | 49.3  | 53.1  | 58.9  | 58.5  | 56.7  | 52.2  | 49.8  | 48.1 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 49.1  | 48.2  | 48.2  | 48.3  | 56.9  | 57.3  | 64.5  | 62.1  | 61.1  | 56.9  | 53.6  | 51.4 |
|                                                                                  |             | MCDB   | 49.2  | 48.5  | 48.8  | 48.9  | 60.9  | 58.6  | 67.1  | 63.6  | 62.1  | 58.0  | 54.1  | 51.9 |
|                                                                                  | 2%          | WB     | 47.5  | 46.6  | 46.8  | 47.3  | 54.2  | 55.9  | 62.0  | 61.3  | 59.8  | 55.6  | 52.2  | 50.1 |
|                                                                                  |             | MCDB   | 48.0  | 47.1  | 47.4  | 48.3  | 57.0  | 57.2  | 64.4  | 62.7  | 61.1  | 56.9  | 52.8  | 50.7 |
|                                                                                  | 5%          | WB     | 46.6  | 45.7  | 46.1  | 46.5  | 51.8  | 55.1  | 60.7  | 60.1  | 58.7  | 54.2  | 51.4  | 48.9 |
|                                                                                  |             | MCDB   | 47.4  | 46.4  | 46.7  | 47.8  | 54.2  | 56.5  | 62.4  | 61.5  | 60.1  | 55.4  | 52.1  | 49.5 |
|                                                                                  | 10%         | WB     | 45.5  | 44.8  | 44.9  | 45.8  | 50.3  | 54.4  | 59.7  | 59.4  | 57.4  | 53.1  | 50.3  | 47.7 |
|                                                                                  |             | MCDB   | 46.6  | 45.8  | 45.7  | 47.0  | 51.7  | 56.1  | 61.4  | 60.9  | 58.9  | 54.4  | 51.0  | 48.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.3   | 4.1   | 3.7   | 3.7   | 3.7   | 3.1   | 3.4   | 3.1   | 3.2   | 3.3   | 3.9   | 4.5  |
|                                                                                  |             | MCDBR  | 4.7   | 4.7   | 4.1   | 4.4   | 6.5   | 4.2   | 4.4   | 3.4   | 3.6   | 4.0   | 4.3   | 5.0  |
|                                                                                  |             | MCWBR  | 6.4   | 5.1   | 4.5   | 3.6   | 4.5   | 3.5   | 4.2   | 4.1   | 4.2   | 4.9   | 5.9   | 6.3  |
|                                                                                  | 5% WB       | MCDBR  | 4.5   | 4.4   | 3.7   | 4.0   | 6.1   | 3.7   | 4.0   | 3.1   | 3.4   | 3.9   | 4.0   | 4.7  |
|                                                                                  |             | MCWBR  | 6.3   | 5.2   | 4.6   | 4.1   | 5.0   | 3.6   | 4.0   | 4.5   | 4.6   | 5.3   | 6.0   | 6.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.336  | 0.353 | 0.369 | 0.387 | 0.372 | 0.373 | 0.384 | 0.382 | 0.369 | 0.352 | 0.332 | 0.309 |      |
|                                                                                  | taud        | 2.145  | 2.243 | 2.297 | 2.295 | 2.413 | 2.443 | 2.426 | 2.433 | 2.443 | 2.346 | 2.208 | 2.112 |      |
|                                                                                  | Ebn at Noon | 146    | 208   | 243   | 259   | 273   | 274   | 268   | 259   | 244   | 208   | 148   | 115   |      |
|                                                                                  | Edh at Noon | 17     | 24    | 30    | 35    | 33    | 33    | 33    | 30    | 26    | 22    | 16    | 12    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 102    | 293   | 688   | 1222  | 1703  | 1697  | 1573  | 1269  | 821   | 403   | 145   | 67    |      |
|                                                                                  | RadStd      | 12     | 43    | 49    | 109   | 121   | 137   | 144   | 86    | 71    | 26    | 15    | 8     |      |

**Historical Trends**

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (7 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page